Executive Summary

01

New customers exhibit the largest churn rates, suggesting onboarding is a critical stage in the customer journey.

02

Customers with higher complaint volumes are substantially more likely to leave the bank.

03

Low mobile-app engagement is one of the strongest predictors of churn, with low-engagement customers approximately six times more likely to churn.

04

Customers holding multiple products demonstrate stronger retention than single-product customers.

Project Objective

This analysis was undertaken to identify the key drivers of customer churn within a UK retail banking portfolio, and develop actionable recommendations to improve customer retention.

The project utilised:

- A customer dataset containing 5,000 individual customers
- Twelve months of transactional and behavioural activity data, including spending, account balances, mobile engagement, branch visits, complaints and product holdings.
- Customer demographic information including age, income, region and tenure.

The objective was to understand which customer segments exhibit the highest churn risk, identify behavioural indicators associated with attrition, and provide data-driven recommendations to strengthen customer retention and long-term profitability.

Data Quality & Limitations

- Analysis based on 5,000 customers and 12 months of activity.
- Data cleaning and standardisation performed to address missing values, duplicate records and inconsistent formatting.
- Findings identify associations with churn and should not be interpreted as proof of causation.
- Additional variables such as customer satisfaction, marketing activity and competitor behaviour were unavailable.

Methodology

Tools used: SQL | Python (Pandas) | Tableau

01

Data Quality Assessment

Reviewed customer and activity datasets to identify missing values, duplicate records, inconsistent formatting, and potential data quality issues that could impact analysis.

02

Data Cleaning & Standardisation

Cleaned and standardised key fields including income, age, region, churn status, and activity data. Removed exact duplicate records, resolved conflicting duplicates and applied validation rules to improve data reliability.

03

Customer-Level Dataset Construction

Aggregated 12 months of customer activity data and combined it with customer demographic information to create a single analytical dataset containing customer characteristics, behaviours, and churn outcomes.

04

Exploratory Data Analysis

Analysed distributions, trends and relationships across customer demographics, tenure, engagement, complaints, product holdings, and financial activity to better understand the customer base.

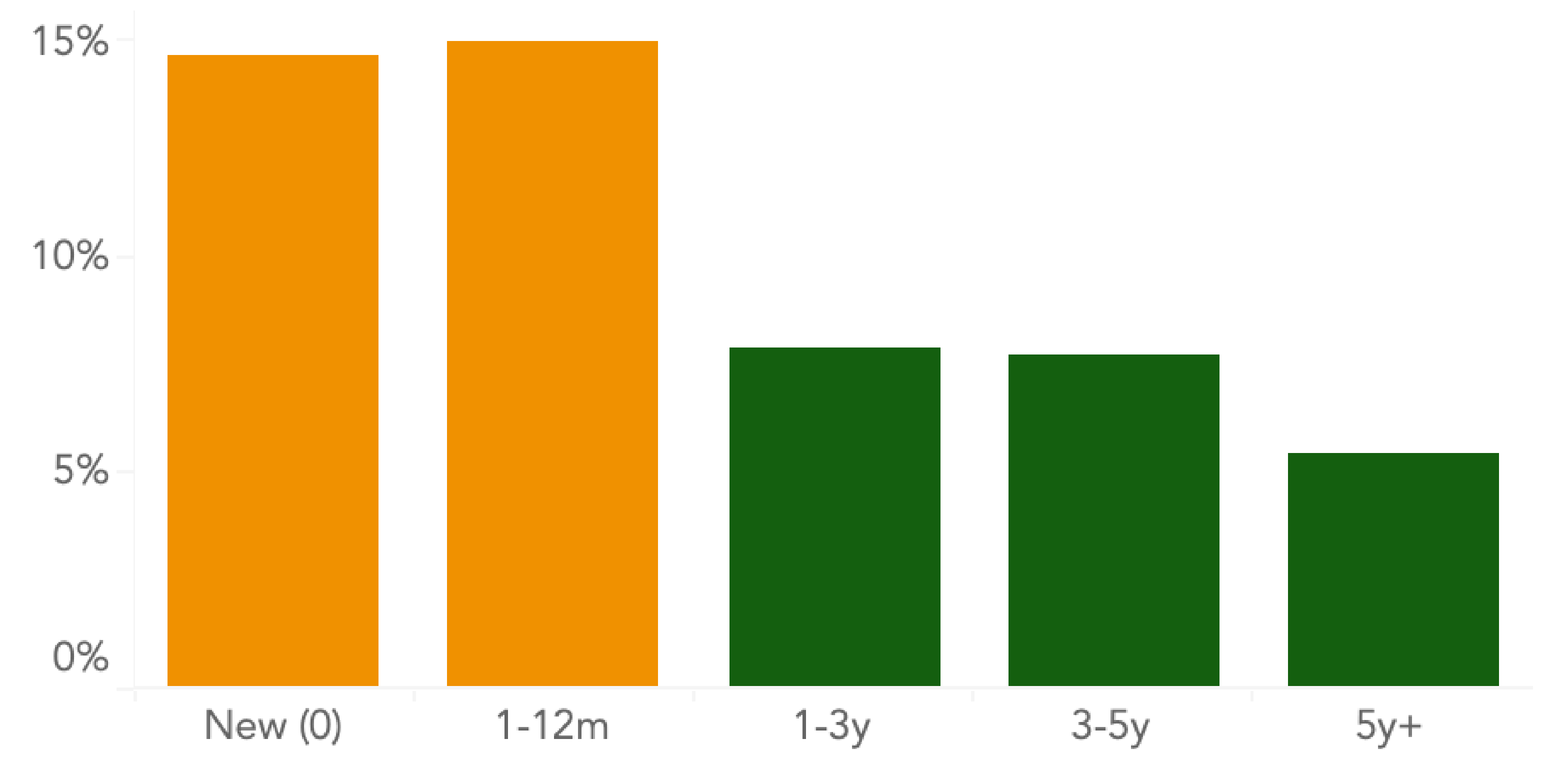
05

Churn Driver Identification

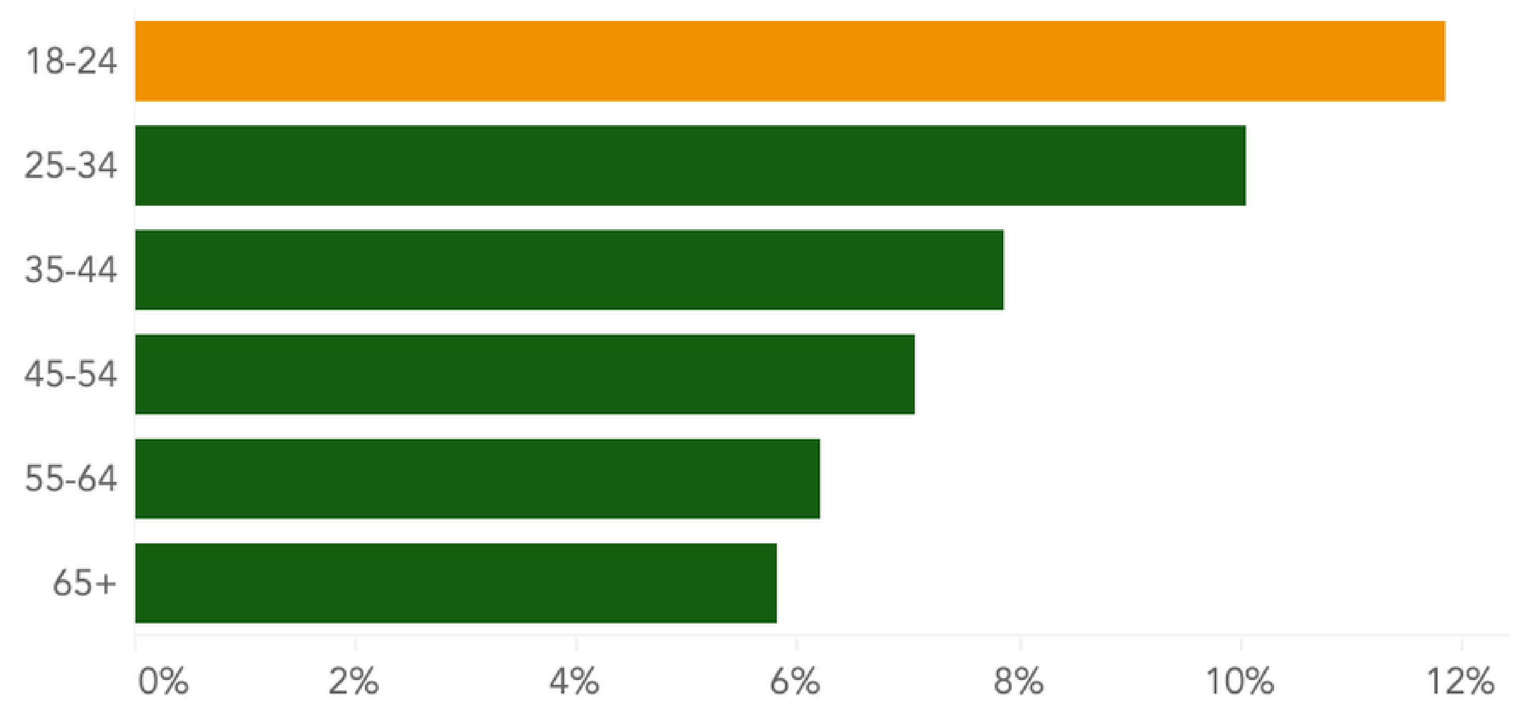
Segmented customers by age, tenure, engagement, complaints, income, and product holdings to identify characteristics associated with higher churn rates and develop targeted retention recommendations.

Finding 1:
New Customers Drive the Bank's Highest Churn Risk

Newer customers exhibit the highest churn rates



Younger customers exhibit the highest churn rates

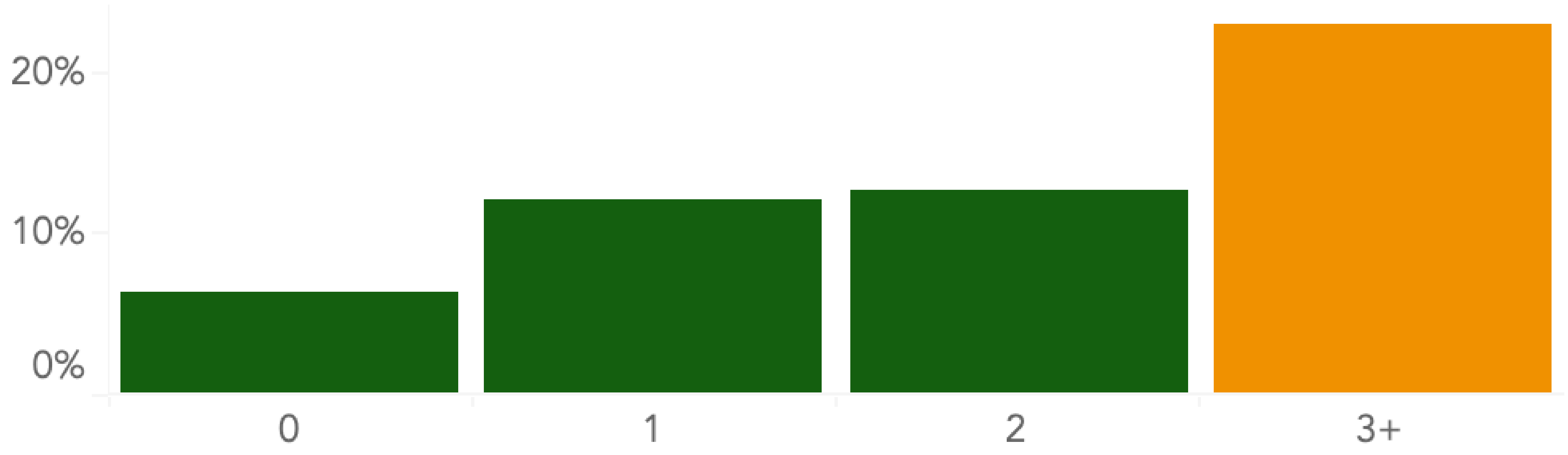


- **18-24** year old customers churn the most.
- **25.4%** of customers aged 18-24 are brand new.
- 0-month tenure customers churn at **14.6%**.
- Strong relationship between age and tenure (**r = 0.67**).
- Early-stage customers represent the highest-risk segment.

- Results suggest elevated churn among younger customers may be partly explained by their higher concentration of new-to-bank customers
- **Recommendation: improve onboarding and early engagement programmes for newly acquired customers, particularly younger customers entering the banking system for the first time.**

Finding 2:
Complaint Volume is Strongly Associated With Customer Churn

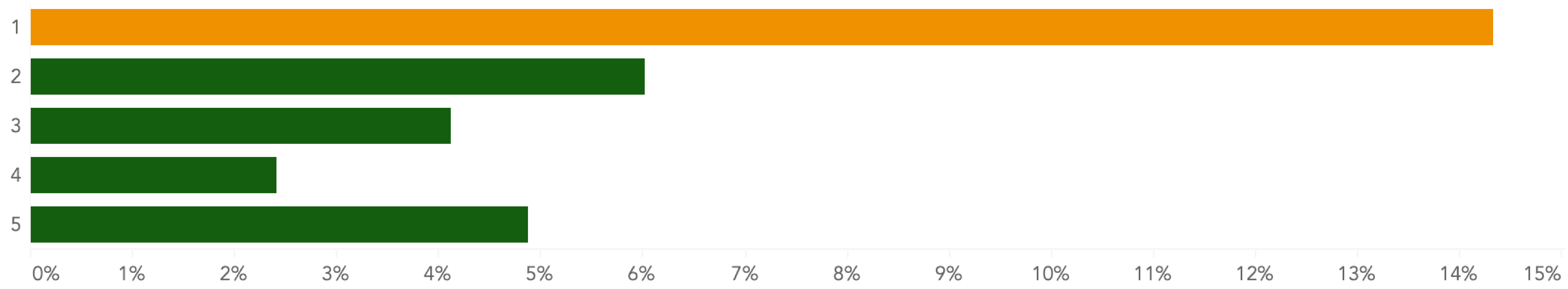
More complaints = more churn



- Customers with multiple complaints are significantly more likely to churn. Service issues appear to be a major driver of attrition.
- **Recommendation: prioritise complaint resolution and service recovery.**

Finding 3:
Deeper Customer Relationships are Associated with Higher Retention

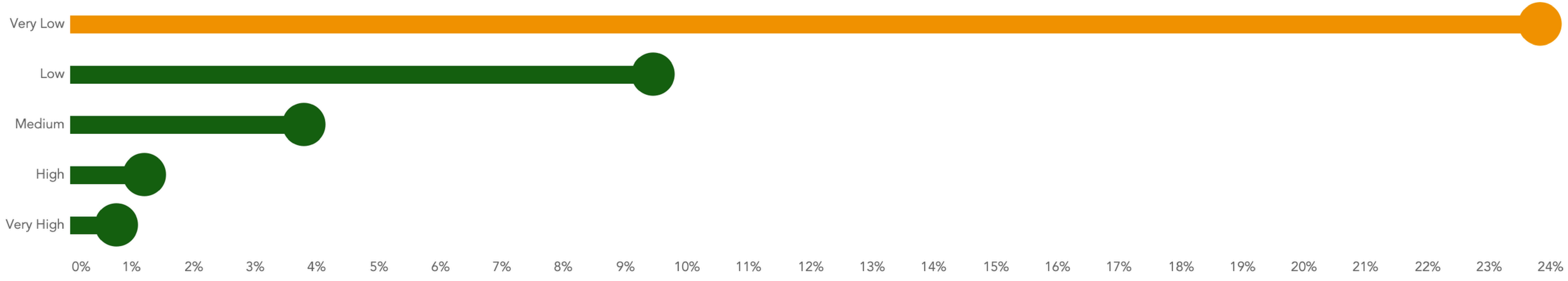
Single-product customers churn more



- Single-product customers exhibit the highest churn rates.
- Multi-product customers have longer tenure and lower churn.
- **Recommendation: cross-sell relevant products to eligible customers.**

Finding 4:
Low Digital Engagement is the Strongest Predictor of Churn

Low-engagement customers are 6x more likely to churn



- Low mobile engagement customers are **6x more likely to churn** than customers with medium engagement.
- Branch visits show little relationship with churn meaning that digital engagement appears significantly more important than physical branch usage.
- **Recommendation: increase app engagement through rewards, personalised offers and proactive communication.**

Prioritised Retention Actions

01

Improve onboarding and early engagement programmes for newly acquired customers, particularly younger customers entering the banking system for the first time.

02

Prioritise complaint resolution and service recovery.

03

Cross-sell relevant products to eligible customers.

04

Increase app engagement through rewards, personalised offers and proactive communication.